UNIVERSITY of NICOSIA | DEPARTMENT OF COMPUTER SCIENCE

Department of Computer Science
COMP–431 Computer Security
Semester Project
100 points possible (7% of final grade)
Due Date: Week 12

General Description

Students will work in teams of 3–4 members to design, develop, and evaluate a practical computer security solution that applies the concepts and techniques studied in this course. The project must involve the implementation or simulation of a security mechanism, tool, or system, rather than a purely theoretical or literature-based study. Projects should address a realistic security problem and demonstrate the use of appropriate techniques such as cryptography, authentication and access control, malware analysis, secure coding, intrusion detection, firewalls, or layered defense strategies. Each team is expected to produce a working prototype, conduct testing or attack simulations, and evaluate the effectiveness of their solution. Emphasis will be placed on correct design, security awareness, technical depth, and clear presentation of results.

Project Requirements

Each team must:

- Work in groups of 3-4 students
- Design and implement a functional prototype or simulator

- Incorporate at least two security mechanisms or techniques covered in the course (unless permission is granted by the instructor)
- Perform testing, evaluation, or simulated attacks (depending on the topic)
- Submit all required deliverables on time

Deliverables and Timeline

1. Project Proposal (End of Week 3) - short document (1-2 pages) including project title and description, problem statement and objectives, proposed approach, technologies/tools to be used, team member responsibilities.
2. In-Class Presentation/Demo (Week 12) - 15 minute presentation, live demo or simulation, discussion of security mechanisms and evaluation results.
3. Final Report (End of Study Week) - technical report (max 10 pages) system design and architecture, implemented security mechanisms, threat model and risk analysis, testing/attack simulations, results and discussion, challenges and lessons learned

Note: Students who opt out from the presentation will receive 0 credit for the entire project.

Suggested Topics

See course site

Prepared by Prof Dionysiou